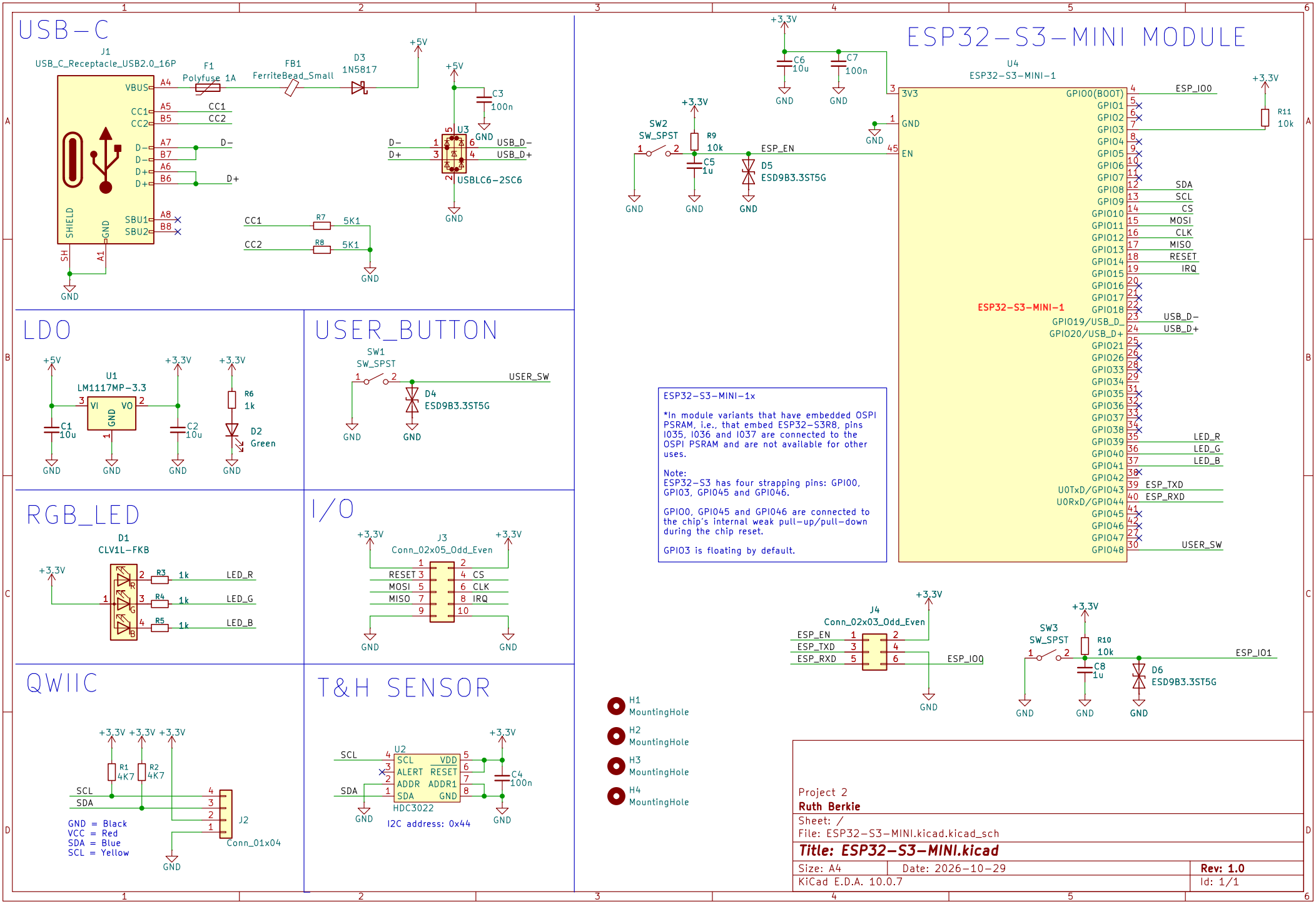

USB-C
J1
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
D-
D+
SHIELD
GND
SBU1
SBU2
SH
A1
F1
Polyfuse 1A
FB1
FerriteBead_Small
D3
1N5817
+5V
C3
100n
U3
USBLC6-2SC6
USB_D-
USB_D+
R7
5K1
R8
5K1
LDO
U1
LM1117MP-3.3
VI
VO
C1
10u
C2
10u
+3.3V
R6
1k
D2
Green
USER_BUTTON
SW1
SW_SPST
USER_SW
D4
ESD9B3.3ST5G
RGB_LED
D1
CLV1L-FKB
R3
1k
R4
1k
R5
1k
LED_R
LED_G
LED_B
I/O
J3
Conn_02x05_Odd_Even
RESET
MOSI
MISO
CS
CLK
IRQ
QWIIC
R1
4K7
R2
4K7
SCL
SDA
GND = Black
VCC = Red
SDA = Blue
SCL = Yellow
J2
Conn_01x04
T&H SENSOR
U2
HDC3022
I2C address: 0x44
C4
100n
ESP32-S3-MINI MODULE
U4
ESP32-S3-MINI-1
3V3
EN
GPIO0(BOOT)
ESP_IO0
SDA
SCL
CS
MOSI
CLK
MISO
RESET
IRQ
LED_R
LED_G
LED_B
ESP_TXD
ESP_RXD
USER_SW
USB_D-
USB_D+
C6
10u
C7
100n
SW2
SW_SPST
R9
10k
C5
1u
D5
ESD9B3.3ST5G
ESP_EN
R11
10k
ESP32-S3-MINI-1x
*In module variants that have embedded OSPI PSRAM, i.e., that embed ESP32-S3R8, pins 1035, 1036 and 1037 are connected to the OSPI PSRAM and are not available for other uses.
Note:
ESP32-S3 has four strapping pins: GPIO0, GPIO3, GPIO45 and GPIO46.
GPIO0, GPIO45 and GPIO46 are connected to the chip's internal weak pull-up/pull-down during the chip reset.
GPIO3 is floating by default.
J4
Conn_02x03_Odd_Even
ESP_IO0
SW3
SW_SPST
R10
10k
C8
1u
D6
ESD9B3.3ST5G
ESP_IO1
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
Project 2
Ruth Berkie
Sheet: /
File: ESP32-S3-MINI.kicad.kicad_sch
Title: ESP32-S3-MINI.kicad
Size: A4
Date: 2026-10-29
Rev: 1.0
KiCad E.D.A. 10.0.7
Id: 1/1